Data-Driven Forecasting & Peak Demand Prediction for Afficionado Coffee Roasters

An exploratory analysis of hourly demand patterns and preliminary machine-learning evaluation

Research paper draft | Prepared from the project's recorded analysis outputs

Abstract

This study examines transaction-level sales records from Afficionado Coffee Roasters to characterize recorded demand by hour, store, and product category, identify peak periods, and assess a preliminary machine-learning model for aggregated store-hour demand. The dataset contains 149,116 transaction records and 214,470 units sold, with calculated revenue of $698,812.33. The highest recorded hour by both quantity and calculated revenue is 10:00, accounting for 26,713 units and $88,673.39 in revenue. A Random Forest Regressor was evaluated on a random holdout of aggregated store-hour observations. The modeling table contains only 43 rows, with 11 test observations; reported mean absolute error (MAE) is 682.654, root mean squared error (RMSE) is 1,080.529, and R^2 is 0.263. These results indicate limited preliminary explanatory/predictive performance on the constructed aggregate dataset. Crucially, the source data contains transaction time but no complete transaction-date field. Consequently, it cannot support genuine daily or multiweek forecasts, chronological validation, or claims of reliable future-date prediction. The project is therefore best interpreted as an exploratory hourly demand-pattern analysis and preliminary modeling exercise. Future work should obtain date-stamped transaction data and use chronological evaluation.

Keywords

Coffee retail analytics; demand patterns; peak-hour analysis; exploratory data analysis; Random Forest Regressor; retail operations; forecasting limitations.

1. Introduction

Demand analysis can help retail food and beverage businesses understand when sales are concentrated and how demand differs across stores and product categories. Such information can inform staffing discussions, inventory planning, and further data collection. However, operational forecasting requires a dataset with a valid time sequence and evaluation that reflects the intended prediction task. A model fitted to aggregated hour-of-day observations is not equivalent to a model forecasting future calendar dates.

1.1 Objectives

The project has four objectives: (1) summarize the available transaction data and assess basic data quality; (2) identify recorded peak-demand hours using quantity and calculated revenue; (3) compare demand across stores and product categories; and (4) evaluate a preliminary Random Forest regression model for aggregated store-hour demand while documenting the limits of the available data.

2. Dataset and Data Preparation

The supplied spreadsheet contains 149,116 transaction records and 11 fields: transaction_id, year, transaction_time, transaction_qty, store_id, store_location, product_id, unit_price, product_category, product_type, and product_detail. The observed year is 2025, and transaction times range approximately from 06:00 to 20:59. The available quality checks reported zero duplicate rows and zero missing values in the inspected dataset. Total quantity is 214,470 units. Revenue was calculated as transaction quantity multiplied by unit price, producing total calculated revenue of approximately $698,812.33.

2.1 Critical time-field limitation

Although a year field and time-of-day field are present, the source does not include a complete transaction date such as YYYY-MM-DD. Therefore, records cannot reliably be ordered into successive calendar days. The analysis can describe the distribution of transactions by time of day, but cannot reconstruct daily demand, weekday/weekend effects, seasonal trends, or

consecutive hourly lags. This limitation governs both model design and interpretation.

3. Methodology

The workflow includes data inspection and quality reporting, descriptive aggregation, visualization, feature preparation, and preliminary regression evaluation. Records are summarized by transaction hour, store, and product category. Revenue is derived from quantity multiplied by unit price. Visualizations and CSV reports are generated for hourly demand, store and category comparisons, and model diagnostics.

3.1 Preliminary regression model

A RandomForestRegressor was trained to estimate total_quantity for aggregated store-hour rows using transaction-hour and store-related features. The resulting modeling dataset contains 43 rows, of which 32 were used for training and 11 for testing. The evaluation used a random holdout split. This split evaluates performance on held-out aggregate rows from the constructed table; it does not represent a chronological test on future dates. The reported feature importance assigns the largest share to transaction_hour (0.9361), followed by smaller store-related contributions, including store_location_Astoria (0.0188), store_id_3 (0.0157), store_id_8 (0.0099), and store_location_Lower Manhattan (0.0070). Feature importance is model-specific and should not be interpreted as causal evidence.

4. Results

The analysis covers 149,116 transaction records, 214,470 units sold, and $698,812.33 in calculated revenue. The peak recorded hour by quantity is 10:00, with 26,713 units. The peak recorded hour by revenue is also 10:00, with $88,673.39. These findings indicate that the morning period around 10:00 is a prominent time-of-day demand peak in the supplied records. Store- and category-level reports provide additional descriptive comparisons; conclusions from those comparisons should be based on the accompanying generated tables and figures.

4.1 Model evaluation

The Random Forest regression model produced MAE = 682.654, RMSE = 1,080.529, and R^2 = 0.263 on 11 test observations. Median absolute error was 365.718, and maximum absolute error was 2,865.812. The positive but modest R^2 indicates that the model explains only a limited share of the variation in the test sample under this particular random split. The small number of aggregate rows and test observations means these metrics are unstable and should be treated as preliminary rather than as robust evidence of deployment performance.

Table 1. Summary of principal results

Measure	Result
Dataset records	149,116
Total units sold	214,470
Calculated revenue	$698,812.33
Peak quantity hour	10:00 (26,713 units)
Peak revenue hour	10:00 ($88,673.39)
Model	RandomForestRegressor
Aggregated modeling rows	43
Test rows	11
MAE	682.654
RMSE	1,080.529

Measure	Result
R^2	0.263

5. Discussion

The descriptive results identify a clear recorded peak at 10:00 for both quantity and calculated revenue. This may be useful as an initial operational signal, for example when investigating morning staffing or replenishment practices. It should not be assumed that the same peak occurs every day or throughout the year because the available dataset does not retain full transaction dates. Similarly, the preliminary regression results should not be used directly to schedule future inventory or staffing: the model was trained on a very small aggregated table and was not evaluated on future calendar periods.

6. Limitations

The main limitation is the absence of a complete transaction date. This prevents genuine daily, weekly, and multiweek forecasting, chronological train/test splitting, and construction of valid calendar-based lag features such as previous-day or previous-week demand. A second limitation is the small modeling sample: 43 aggregated store-hour rows with only 11 test observations. A random split may also place related store-hour patterns in both training and test subsets, making the result different from a true future-period evaluation. The calculated revenue metric assumes the supplied unit_price and transaction_qty are appropriate for this calculation; no independent reconciliation with accounting records was performed. The analysis is observational and does not establish causal effects.

7. Recommendations and Future Work

Future work should obtain transaction-level records with a complete date and timestamp, consistent store identifiers, product details, quantity, price, and any relevant promotion or closure indicators. After validation and cleaning, data should be aggregated into a continuous hourly or daily series with explicit treatment of missing intervals. Models should be compared against simple baselines and suitable time-series or machine-learning alternatives. Evaluation should use chronological splits or rolling-origin validation and report error metrics for multiple forecast horizons. If adequate history becomes available, calendar effects, holidays, promotions, weather, and stock availability may be considered where reliable data can be obtained. Until then, the dashboard should be used for descriptive exploration rather than automated future-demand decisions.

8. Conclusion

This project provides an exploratory view of Afficionado Coffee Roasters transaction patterns and a preliminary regression exercise. In the supplied data, 10:00 is the peak recorded hour for both quantity and calculated revenue. The Random Forest model achieves MAE of 682.654, RMSE of 1,080.529, and R^2 of 0.263 on a small random holdout of aggregated store-hour rows. These metrics are not evidence of reliable future-date forecasting. The absence of a complete transaction date is a fundamental constraint. The most important next step is to obtain date-stamped historical data and evaluate forecasting methods on chronological holdouts before operational use.

References

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Appendix A. Reproducibility and project outputs

The project scripts produce data-quality summaries, hourly demand summaries, store and category comparisons, model predictions and metrics, feature-importance tables, and diagnostic figures. The dashboard presents these outputs interactively. For reproducibility, the final repository should include the source scripts, dependency list, documentation, and generated reports/figures. The source spreadsheet should only be published if its licensing and assignment terms permit redistribution.